

VSDS Mahesh Akavarapu

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Research Interests

Computational Linguistics / Natural Language Processing, Historical Linguistics
 Specifically: Computational Historical Linguistics, Speech Processing, Low Resource NLP

Education

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|------------|---|-----------------------|
| PhD | Indian Institute of Technology Kanpur , Computer Science
• GPA: 3.84/4.00 (9.6/10.0)
• Advisor: Prof. Arnab Bhattacharya | July 2019 – Sept 2024 |
| BT | Indian Institute of Technology Kanpur , Computer Science, Physics (Second Major)
• GPA: 3.72/4.00 (9.3/10.0) | July 2014 – May 2019 |

Awards / Honors

- Prime Minister's Research Fellowship (2019-24):** Grant of INR 200,000 per year (approx. USD 12,000 in total) from Ministry of Education, Government of India
- Academic Excellence Award from IIT Kanpur (2015):** Awarded to Top 10% students of the batch

Experience

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| University of Tübingen , Postdoctoral Researcher
• Advisor: Prof. Gerhard Jäger | Tübingen, Germany
Jan 2025 – present
1 year 9 months |
| Indian Institute of Technology Kanpur , Research Fellow
• Under Prime Minister's Research Fellowship (Advisor: Prof. Arnab Bhattacharya) | Kanpur, India
Aug 2019 – July 2024
5 years |
| Indian Institute of Technology Kanpur , Student Researcher, Intern
• High Performance Computing in CUDA (Advisor: Prof. Mahendra K. Verma) | Kanpur, India
May 2018 – July 2018
3 months |

Publications

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|---|-----------|
| Hard to Be Heard: Phoneme-Level ASR Analysis of Phonologically Complex, Low-Resource Endangered Languages
V.S.D.S.Mahesh Akavarapu, Michael Daniel, Gerhard Jäger
(Findings of ACL 2026 , San Diego, CA-US) | July 2026 |
| A Case Study of Cross-Lingual Zero-Shot Generalization for Classical Languages in LLMs
V.S.D.S.Mahesh Akavarapu, Hrishikesh Terdalkar, Prमित Bhattacharyya, Shubhangi Agarwal, Vishakha Deulgaonkar, Pralay Manna, Chaitali Dangarikar, Arnab Bhattacharya
(Findings of ACL 2025 , Vienna, Austria) | July 2025 |
| A Likelihood Ratio Test of Genetic Relationship among Languages
V.S.D.S.Mahesh Akavarapu, Arnab Bhattacharya
(Proc. of NAACL 2024 , Mexico City, Mexico) | June 2024 |

Automated Cognate Detection as a Supervised Link Prediction Task with Cognate Transformer

Mar 2024

V.S.D.S.Mahesh Akavarapu, Arnab Bhattacharya
(Proc. of **EACL 2024**, St. Julian's, Malta)

Cognate Transformer for Automated Phonological Reconstruction and Cognate Reflex Prediction

Dec 2023

V.S.D.S.Mahesh Akavarapu, Arnab Bhattacharya
(Proc. of **EMNLP 2023**, Singapore)

Creation of a Digital Rig Vedic Index (Anukramani) for Computational Linguistic Tasks

Jan 2023

V.S.D.S.Mahesh Akavarapu, Arnab Bhattacharya
(Proc. of **World Sanskrit Conference 2023**, Canberra, Australia)

Invited Talks

- Evolutionary Biology-Inspired Language Models in Historical Linguistics, University of Tübingen (Online), Apr. 2024
- Tutorial on LLMs: Finetuning and Prompting, with Arnab Bhattacharya and Shubham K. Nigam, NIT Warangal (Online), Mar. 2024
- Tutorial on Legal Named Entity Recognition, with Shubham K. Nigam, NIT Trichy (Online), Dec. 2022

Teaching

- *Automatic Speech Processing*, Uni. Tübingen, (Summer 2026)

Tutoring

- *Signals and Systems*, CSJM University, Kanpur (Summer 2024)
- (CS 63) *Theory of Computation*, NPTEL, Online (Fall 2023, Fall 2022)
- *Introduction to Machine Learning with Python*, CSJM University, Kanpur (Fall 2022)

Teaching Assistance

- (CS 689) *Computational Linguistics for Indian Languages*, IIT Kanpur (Spring 2024, Fall 2022)
- (CS 315) *Principles of Database Systems*, IIT Kanpur (Spring 2023, Spring 2021)
- (CS 657) *Information Retrieval*, IIT Kanpur (Spring 2022)
- (CS 771) *Introduction to Machine Learning*, IIT Kanpur (Fall 2021)
- (CS 685) *Data Mining*, IIT Kanpur (Fall 2020)
- (ESC 101) *Fundamentals of Computing*, IIT Kanpur (Spring 2020, Fall 2019)

Technologies

Languages: Python, C/C++, HTML, Shell Scripting etc.

Libraries: PyTorch, TensorFlow, LangChain, HuggingFace etc.

Personal Details

Full Name: Akavarapu, Venkata Satya Durga Sai Mahesh

Gender: Male, **Age:** 29, **Nationality:** Indian